About breaking your storyteller's spirit by applying algebraic theory to turn amnesiac in the best info role in Blood of the clocktower. And related shennanigans

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Abstract

We take a look at the game *Blood on the Clocktower*. We demonstrate that the *Storyteller* can always unambiguously answer the question

Is my ability the value of our canonical amnesiac mapping for player K?

and that the answer is truth-functionally equivalent to the hot-or-cold question *"Is player K the Demon?"*. With this question we can therefore effectively turn every amnesiac into the strongest info role.

1 Preliminaries

We assume that you, dear reader, are familiar with *Blood on the Clocktower* (*BotC*) including the basic rules, factions and goals. We will therefore only recall *BotC* basics that should be common knowledge, but might not be.

Definition 1.1 (Storyteller)**.** A *Storyteller* is a member of the gaming session that acts as a neutral moderator, who's sole purpose it is, to make innocent players suffer in confusion.

Definition 1.2 (Players' hidden agenda)**.** A player's hidden agenda is to make the *Storyteller* suffer by playing as quirky, weird and time-consuming as possible.

Definition 1.3 (Notations)**.** For all following pages, if not otherwise specified:
Let P be set of players in a game G of *BotC* and P_G be the cyclic graph of their seating.
$N = |P|$. $N = 11$ means a game of eleven players plus storyteller.
$P, K, D \in P$ and are players, where D is the *Player* that drew the demon token.

2 Gossip min-mining

As motivation for our amnesiac approach, we want to look at other question strategies.

Various characters allow *Player*s to ask the *Storyteller* questions. A straightforward approach is to narrow down demon candidates with our question. A common approach is to bisect the player set P with questions such as

Is player X in the half of players that have the demon at their center?

or

Is the demon above the median age?

What other questions can you think of?

Question 2.1. Let's discuss the following question

> *We give each player a color. Starting with the demon and continuing with the demon's next neighbor left, right, left right, we color half the number of players (rounded down) orange. The remaining players are colored green. Is player X orange?*

or in a less formal, more confusing and really boring way.

> *Is player X in the half of players that have the demon at their center?*

Observation 2.2. Let's look at two coloring examples for even and odd cases to

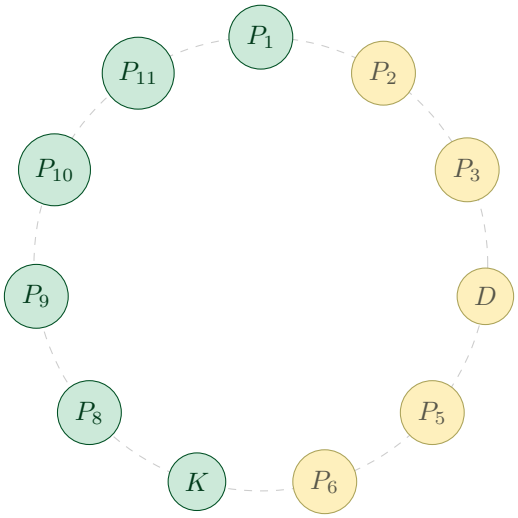


Figure 1: Coloring of odd player count

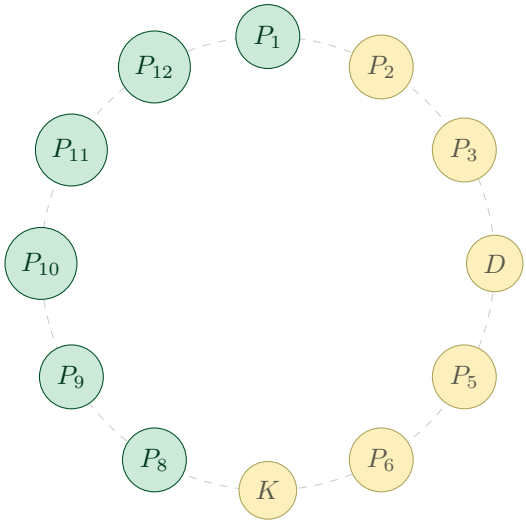


Figure 2: Coloring of even player count

First, we observe that we know that either K or the *Player* (s) seated directly across from K are green and therefore not the demon.

For N *is odd*, we learn two players that are not the demon

Let's study $N = 11$

Claim 2.3. Q2.1 clears exactly one player after one question, performs as good as any binary search after two or more questions and annoys the *Storyteller* nicely

Proof. We leave the proof as a nice exercise for you, dear reader. □

Remark 2.4 (Ternary questions). The rulebook defines questions as ternary over the answer set yes, no, i don't know. As some redditor showed, we can apply advanced question design to turn an "I don't know" in a "We know more"

References